

PRIYANKA B

Data Engineer

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Professional Summary

Results-driven Analytic Data Engineer with 9+ years of experience in designing, implementing, and optimizing Big Data and Cloud-based solutions across AWS, Azure, and GCP platforms. Proven expertise in data ingestion, modeling, querying, processing, and analysis, with a strong focus on enterprise-level data integration and real-time analytics.

Expert in Big Data technologies including Hadoop, Spark, Hive, Kafka, and HDFS, with hands-on experience in building scalable data pipelines using Apache Spark, Spark-Scala, Apache Flume, Sqoop, PIG, Spark-SQL, Hue, and Trifacta. Skilled in leveraging Hadoop distributions such as IBM Big Insights, Hortonworks, Cloudera, and AWS EMR to support data engineering workflows.

Proficient in Cloud Data Engineering with extensive experience in Azure Data Factory, Azure Synapse Analytics, Azure Data Lake, Azure Databricks, Azure Cosmos DB, and AWS services such as Redshift, Glue, S3, Lambda, EC2, EMR, Kinesis, and DynamoDB. Strong knowledge of GCP services, including BigQuery, Cloud Functions, Data Proc, Pub/Sub, and Cloud Composer.

Well-versed in real-time data streaming and event-driven architectures using Kafka, Spark Streaming, and Flume, developing data pipelines that facilitate high-volume data processing and analytics. Skilled in REST API development using Python for large-scale applications and integrating with cloud-based environments.

Extensive experience in CI/CD pipelines using Jenkins, version control with Git, and release management using Urban Code Deployment (UCD). Strong background in SQL and NoSQL databases, including PostgreSQL, MySQL, Oracle DB, MSSQL Server, and MongoDB. Proficient in Apache Airflow, Control-M, ESP, and D-series for job scheduling and orchestration.

Adept at performance tuning, optimization, and troubleshooting in big data ecosystems, ensuring efficient data warehouse and Hadoop data mart implementations. Experienced in working within Agile environments, collaborating cross-functionally, and following best practices for cloud security, automation, and data governance.

A forward-thinking data engineering professional with a deep passion for data-driven solutions, automation, and cutting-edge cloud technologies, capable of driving enterprise-wide digital transformations and delivering high-impact business insights through scalable and cost-effective data solutions.

Technical Skills

- **Big Data Tools:** Hadoop, Hive, Apache Spark, PySpark, HBase, Kafka, YARN, Sqoop, Impala, Oozie, Pig, Map Reduce, Zookeeper, Flume
- **Hadoop Distributions:** EMR, Cloudera, Hortonworks
- **Cloud AWS:** EC2, S3, EMR, RDS, Glue, Presto, Lambda, RedShift, Azure - Data Services
- **GCP:** Cloud Storage, Big Query, Compute Engine, Cloud Composer, Data Proc, Data Flow, Pub-sub, GCP Looker
- **Azure:** Azure Data Lake, Data Bricks, Azure Data Factory, Azure Monitoring, Active Directory, Synapse, Key Vault, SQL Azure
- **BI and Data Visualizations:** GCP Looker, ETL -Informatica, SSIS, Talend, Tableau, Power BI
- **Relational Databases:** Oracle, SQL Server, Teradata, MySQL, PostgreSQL, Netezza
- **No SQL Databases:** Cassandra, MongoDB, HBase
- **Programming Languages:** Scala, Python, R, Java
- **Scripting:** Python, Shell scripting
- **Build Tools:** Apache Maven, SBT, Jenkins, Bitbucket
- **Version Controls:** GIT, SVN
- **Operating Systems:** Unix, Linux, Mac OS, CentOS, Ubuntu, Windows
- **Tools:** PUTTY, Putty-Gen, Eclipse, IntelliJ, Toad

Professional Experience

Wells Fargo

GCP Data Engineer

Oct 2023 - Present

Charlotte, NC

- Involved in full Software Development Life Cycle (SDLC) - Business Requirements Analysis, preparation of Technical Design documents, Data Analysis, Logical and Physical database design, Coding, Testing, Implementing, and deploying to business users.
- Proficient in Data Processing, with expertise in creating clusters using Apache Hadoop and Apache Spark on Google Cloud Platform (GCP).
- We are using a Data flow automatic template for managing services GCS TO BIG Query and BIG query to GCS.

- Implemented real-time message subscription using Google Cloud Pub/Sub, facilitating seamless data processing for large datasets during on-premises to Google Cloud Platform migration.
- Worked on Google Cloud Platform (GCP) in all the big data products Big Query, Cloud Data Proc, Google Cloud Storage, and Composer (Air Flow as a service).
- Extensive knowledge and understanding of ETL and ELT for real-time batch processing and streaming processes using GCP.
- Have good experience with GCP tools such as BIG QUERY, GCS COMPOSER, GCS, PUB/SUB, DATA FLOW, DATA PROC, Cloud SQL, GKE, And Airflow.
- We are collecting data from source systems like IBM Data streams such as IoT devices and sensors, Text files, spreadsheets, images, video, and audio recordings, and finally data is moving to GCP Cloud.
- Explore GCP's data processing and streaming services, such as Google Cloud Dataflow or Apache Beam, which are like IBM Streams and can be used to process real-time data.
- Transformed and analyzed the data using Pyspark, HIVE, based on ETL mappings.
- Extensively used spark SQL and Data frames API in building spark applications.
- Developed spark programs created the data frames and worked on transformations and actions.
- Analyzed source data coming from different sources like SQL Server tables, XML files, and Flat files then transformed according to business rules using Informatica and loaded the data into target tables.
- Designed and developed several complex mappings using various transformations like Source Qualifier, Aggregator, Router, Joiner, Union, Expression, Lookup, Filter, Git Hub Update Strategy, Stored Procedure, Sequence Generator, etc.
- Developed Python scripts to automate the ETL process using Apache Airflow and CRON scripts in the UNIX operating system as well.
- Used Apache airflow in the GCP Composer environment to build data pipelines and explored various airflow operations like bash operator, Hadoop operators, and branching operators.
- Used cloud shell SDK in GCP to configure the services Data Proc, Storage, and big query.
- Extensive use of cloud shell SDK in GCP to configure/deploy the services using GCP Big Query.
- Developed ELT processes from the files from Abinitio and Google Sheets in GCP with Compute being data proc (PySpark) and big query.
- Changing the existing Data Models using Erwin for enhancements to the existing Datawarehouse projects.
- Automated infrastructure deployment using Terraform, Kubernetes, and Docker, reducing manual operational overhead.
- Worked on Google Cloud Platform (GCP) services like Compute Engine, cloud load balancing, cloud storage, and cloud SQL.
- Designed and operationalized cloud-based platform components on GCP (BigQuery, DataProc, Dataflow, Cloud Storage, GKE, Pub/Sub) to ensure high availability and scalability.
- Develop and deploy the outcome using Spark and Scala code in the Hadoop cluster running on GCP.
- Experience in Google Cloud components, Google container builders, and GCP client libraries.
- Worked with PaaS services and cloud orchestration, ensuring optimal resource allocation and cost efficiency.
- Experienced with reporting tools, software, and other applications, including SQL Database, Looker, Tableau, Power BI, and Salesforce dashboards (reports designing tool) and Data warehouse.
- Supported various business teams with Data Mining and Reporting by writing complex SQLs using Basic and Advanced SQL including OLAP functions like Ranking, partitioning, and windowing functions.
- Using the g-cloud function with Python to load Data into Big query for arrival CSV files in the GCS bucket.
- We use Looker, which is platform-independent, to connect to data in Google Cloud's Big Query, Data Proc, and other public clouds.
- Environment: Data proc, Looker, Scala, Apache spark, Talend, Google Cloud Platform (GCP), Cloud Storage, Big Query, Compute Engine, Cloud Composer, Data Flow, Pub-sub, Python, MySQL, Teradata, Erwin, EMR, GitHub.

Citrix

Sr. Data Engineer

Jul 2021 - Sep 2023

Fort Lauderdale, FL

- Hands-on major components in Hadoop Echo Systems like Spark, HDFS, HIVE, HBase, Zookeeper, Sqoop, and Oozie.
- Conducting Exploratory data analysis in Jupiter notebooks using Python libraries and sharing the data analysis.
- Developing Sqoop jobs to ingest data from various systems of records into Enterprise Data Lake.
- Development of Spark jobs in PySpark and Spark SQL to run on top of hive tables and create transformed data sets for downstream consumption.
- Ingesting data from various source systems like Oracle, SQL Server, Flat files, and JSONs.
- Performance tuning spark and hive jobs by reading execution plans, DAGs, and Yarn logs.
- Creating generic shell scripts to submit Hadoop and spark jobs on EMR and on-prem edge nodes.
- Writing Complex SparkSQL codes to clean, join, transform, and aggregate the datasets and publish them for the Power BI team to produce operational scorecards.
- Writing custom Python modules for reusable Python code.
- Working on designing the MapReduce and Yarn workflow and writing MapReduce scripts, performance tuning, and debugging.
- Worked on migrating on-prem Hadoop cluster data and data pipelines to AWS Cloud.
- Created multiple proof-of-concepts using PySpark and deployed them on the Yarn cluster, comparing Spark's performance to that of Hive and SQL/Teradata.
- Integrated Teradata Warehouse into the EMR cluster. Developed shell scripts to load data from the Teradata Staging area to the Teradata data mart. Handled Errors & and tuned performance in Teradata queries and utilities.
- Set up databases in GCP using RDS, storage using S3 bucket, and configuring instance backups to S3 bucket. Prototype CI/CD system with GitLab on GKE utilizing Kubernetes and Docker for the runtime environment for the CI/CD systems to build test and deploy.
- Solving appropriate partition and bucketing schemes and making sure correct load policies are employed so data can be stored as per requirements.
- Generates ETL scripts to transform, flatten, and enrich the data from source to target using AWS Glue and created event-driven ETL pipelines with AWS Glue.

- Documenting data flow diagrams and technical logic in confluence.
- Worked on setting up AWS DMS and SNS for data transfer and replication and used SQL on the new AWS Databases like RedShift and Relation Data Services.
- Created external tables with partitions using Hive, AWS Athena, and Redshift.
- Created and deployed Lambda functions in AWS using pre-built AWS Lambda Libraries, as well as Lambda functions in Scala using custom libraries.
- Ensured data quality checks and monitoring automation for robust ETL performance.
- Optimizing and tuning the Redshift environment, enabling queries to perform up to 100x faster for Tableau and SAS Visual Analytics.
- Creating Oozie workflows, Coordinators, and scheduling handshake jobs in Control-M.
- Created Session Beans and Controller Servlets for handling HTTP requests from Talend.
- Using Control-M for orchestration, the workflow will be executed through Airflow, AWS EMR, and Glue to run the jobs and glue catalog to capture the metadata, Lambda to process the source data freshness check, and capture the data quality metrics.
- Built data ingestion pipelines for real-time and batch processing using Kafka, Apache Spark, and GCP Dataflow.
- Working with governance teams to ensure metadata management, data lineage, and technical metadata are correctly updated for each data asset.
- Automated workflow scheduling using Airflow DAGs and built monitoring solutions for proactive issue detection.
- Integrated DevOps best practices, implementing CI/CD pipelines using Jenkins, GitHub Actions, and Kubernetes (GKE) for continuous deployment and testing.
- Developed and optimized complex ETL pipelines (3+ years) with dependency management, File Watchers, APIs, Airflow, and Talend.
- Working with production support teams and administration teams to ensure correct access controls are set up on each hive database.
- Working with the master-feature branch model and committing the code with appropriate comments.
- Attending sprint planning, agile ceremonies, and demoing the work products on a bi-weekly basis.
- Environment: AWS RedShift, AWS EMR, ELK, EC2, ETL, AWS Glue, S3, Athena, Hive, Impala, Lambda, Python, PySpark, SparkSQL, Oracle 11g/12c, Teradata, Jira, Bitbucket, Power BI, Control-M, Airflow, Talend, CI/CD pipelines, GKE, Redshift.

Early Warning's

SR Data Engineer

Feb 2019 - Jun 2021

Scottsdale, AZ

- Experience with Azure Synapse, Stream Analytics, Azure Event Hubs, and Azure Event Grid.
- Experience in Migrating SQL database to Azure Data Lake, Azure Data Lake Analytics, Azure SQL Database, and Azure SQL Data Warehouse.
- Data Ingestion to Azure Services, including Azure Data Lake, Azure Storage, Azure SQL, and Azure DW as well as data processing in Azure Data bricks.
- Worked on the design and development of Informatica mappings, and workflows to load data into the staging area, data warehouse, and data marts in SQL Server and Oracle.
- Using Server Manager, created batches and sessions to transport data at present intervals and on-demand.
- Extracted, converted, and loaded data from various sources to Azure Data Storage Services utilizing Azure Data Factory and T-SQL for data lake analytics.
- Performed data transformations for ML OPs, including adding calculated columns, maintaining relationships, establishing various metrics, merging, and appending queries, changing values, splitting columns, and grouping by Date and Time Column.
- Involved in the Data Migration process using Azure by integrating with the GitHub repository and Jenkins.
- Developed Python scripts to do file validations in Databricks and used ADF to automate the process.
- Developed audit, balancing, and control architecture utilizing SQL DB audit tables to manage the Azure ingestion, transformation, and load processes.
- Created tables in Azure SQL DW for business-related data reporting and visualization.
- Utilized Power BI desktop to generate visualization reports, dashboards, and KPI scorecards.
- Conceived, designed, and implemented ETL solutions with SQL Server Integration Services (SSIS).
- Created Pipelines that were built in Azure Data Factory using Linked Services/Datasets/Pipeline/ to extract, transform, and load data from a variety of sources including Azure SQL, Blob storage, Azure SQL Data warehouse, write-back tool, and reverse.
- Created CI-CD Pipelines using Azure DevOps.
- Worked on various issues on existing Informatica Mappings to Produce correct output.
- Native integration with Azure Active Directory (Azure AD) and other Azure services enables us to build a modern data warehouse and machine learning and real-time analytics solutions.
- Used a blend of Azure Data Factory, T-SQL, Spark SQL, and U-SQL Azure Data Lake Analytics, to gather, convert, and load the data from source systems to Azure Data Storage services.
- Configured Spark Streaming to receive real-time data from Kafka. Used Backpressure to control message queuing in the topic.
- Conducted Root Cause Analysis (RCA) for platform issues, identifying and resolving system bottlenecks.
- Created and monitored notifications for data integration events (success/failure).
- Optimized job performance and cluster utilization, leading to 40% cost savings in cloud resource consumption.
- Collaborated with product managers, scrum masters, and engineers to build Agile processes and documentation projects for retrospectives, the backlog, and meetings.
- Environment: Azure Data Storage, Azure Data Factory, Azure Services, Azure SQL server, Azure data warehouse, MySQL, ETL, Kafka, PowerBI, SQL Database, T-SQL, U-SQL, GitHub, Azure Data Lake, Azure Data bricks, SSIS.

Cap Gemini India Pvt. Ltd.

Data Engineer

Mar 2017 - Nov 2018

Hyderabad

- Work closely with Business Analysts and Product Owners to understand the requirements.
- Used Joins in SPARK for making smaller datasets to large datasets without shuffling data across nodes.

- Developed Spark Streaming jobs using Python to read messages from Kafka.
- Write Scala program for spark transformation in Dataproc.
- Used Spark Streaming to receive real-time data from Kafka and store the stream data to HDFS using Python and NoSQL databases such as HBase and Cassandra.
- Prototyped analysis and joining of customer data using Spark in Scala and processed it to HDFS.
- Implemented Spark in EMR for processing Big Data across our One Lake in AWS System.
- Consumed and processed data from DB2.
- Developed Spark/Scala, Python for regular expression (REGEX) project in the Hadoop/Hive environment with Linux/Windows for big data resources.
- Extensive experience with Big Data technologies (3+ years) including Apache Spark, Hive, Hadoop, Parquet, HDFS, and NoSQL databases.
- Downloaded JSON files from AWS S3 buckets.
- Implemented ETL using AWS RedShift/Glue.
- Developed applications using Spark to implement various aggregation and transformation functions of Spark RDD and Spark SQL.
- Used Scala components to implement the credit line policy based on the conditions applied on spark data frames.
- Data sources are extracted, transformed, and loaded to generate CSV data files with Python programming and SQL queries.
- Designed and implemented data lake architectures, enabling efficient data retrieval and processing at scale.
- Using the g-cloud function with Python to load Data into Big query for arrival CSV files in the GCS bucket.
- Write a program to download a SQL Dump from their equipment maintenance site and then load it in the GCS bucket. On the other side load this SQL dump from the GCS bucket to Mysql (hosted in Google Cloud SQL) and load the Data from Mysql to Big query using Python, Scala, spark, and Data proc.
- Process and load bound and unbound Data from Google pub/subtopic to big query using cloud Dataflow with Python.
- Create firewall rules to access Google Data proc from other machines.
- Environment: Spark, Hive, ETL, Data Stage, Informatica, Talend, AWS S3, EMR, RedShift, EC2, Python, SQL, Nifi, Airflow, Jupiter, Cloudera, Hadoop, Apache Kafka, Apache flume.

CONTUS Support Interactive (P) Ltd.

Aug 2014 - Apr 2017

Data Engineer

Chennai

- Configured AWS Data pipeline, transferring data from S3 to Redshift efficiently. Proficient in using Python for AWS Lambda functions, data processing, and automation tasks.
- Engineered end-to-end data processing pipelines using Databricks on AWS, integrating data from diverse sources such as Amazon S3, databases, and streaming platforms.
- Implemented ETL processes in Databricks notebooks, transforming raw data into clean, structured formats for further analysis.
- Proficient in writing and optimizing SQL scripts and stored procedures to manipulate and retrieve data from AWS databases such as Amazon RDS (Relational Database Service) and Amazon Redshift.
- Integrated AWS SDKs with Java for seamless interaction with AWS services.
- Developed serverless Java applications on AWS Lambda.
- Designed automation scripts for Spark-based analytics, connecting multiple data sources with Amazon S3.
- Used AWS Redshift for ETL, extracting, transforming, and loading data from diverse sources.
- Developed PySpark scripts, enabling rapid data testing and efficient processing of information.
- Crafted ETL jobs with Python and Scala, ensuring alignment with client specifications.
- Implemented Slowly Changing Transformation to maintain historical data integrity in the Data warehouse.
- Proficiently handled AWS components: EC2, S3, Redshift, Athena, Glue, Lambda, DynamoDB, and more.
- Architected end-to-end ETL processes in AWS, while meticulously documenting business workflows.
- Collaborated directly with clients to seamlessly retrieve data from different databases.
- Implemented data modeling techniques (Star schema, Snowflake schema) in PostgreSQL, ensuring structured and efficient data storage.
- Implemented fine-grained access control and role-based security in Snowflake to restrict data access and ensure compliance with data governance policies.
- Worked with Snowflake's Time Travel and Data Sharing features to enable data versioning, historical data analysis, and sharing data securely with external partners.
- Conducted performance tuning and optimization of Snowflake workloads, monitoring and analyzing query execution plans for bottlenecks and improvements.
- Utilized Snowflake's Snowscape web interface for interactive data exploration, visualization, and reporting.
- Actively participated in AWS architectural discussions, database development, and rigorous data validation.
- Offered expert guidance to PySpark ETL platform development, enhancing team productivity.
- Environment: Spark, Hive, ETL, Data Stage, Informatica, Talend, AWS S3, EMR, RedShift, EC2, Python, SQL, Nifi, Airflow, Jupiter, Hadoop, Apache Kafka, Apache flume

Education

Gokaraju Rangaraju Institute of Engineering and Technology

Bachelor of Technology in Computer Science and Engineering

Aug 2010 - Apr 2014

Hyderabad, IN